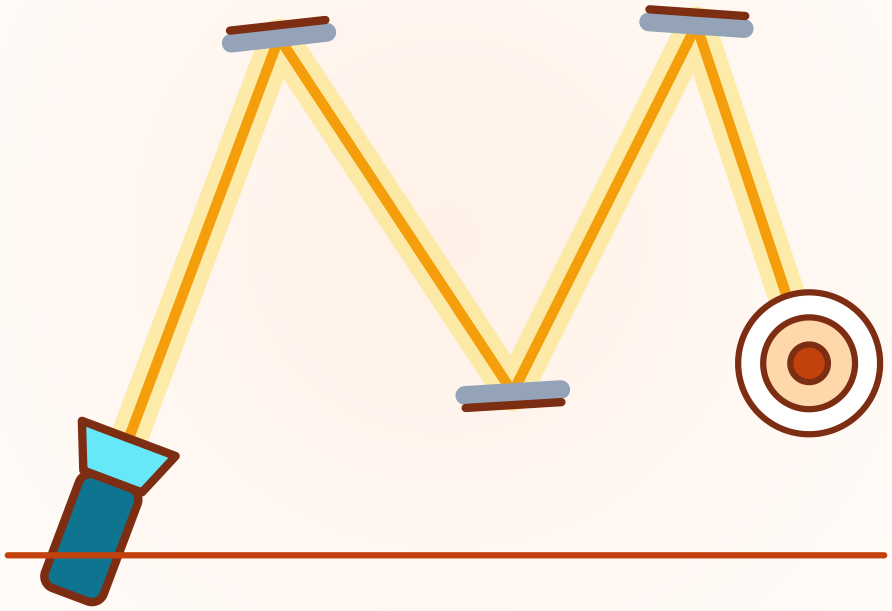
BEAM ME UP

Engineer's Logbook

Plan it. Bounce it. Time it. Hit the target.



ENGINEER ID

Name: _____

Partners: _____

Event: Beam Me Up **Team Leader:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How a light engineer works

Engineers don't guess and hope. They **plan**, **test**, write down what happened, and change **one thing at a time**. Light travels in straight lines until it hits something. Then it can bounce, bend, soak in, pass through, or spread out, and your eye is built to catch it.

1
PLAN
Sketch your mirror path on graph paper before the lights go out.

2
PREDICT
Guess your time and where each holder stands.

3
TEST
Run the course. Three timers, use the middle time.

4
RECORD
Fill a row of the Trial Log right away.

5
IMPROVE
Change one thing. Try again. Graph your best times.

The Engineer's Code

- [ ] Never shine a flashlight or a mirror spot into anyone's eyes.
- [ ] Ask an adult before you turn the lights off.
- [ ] Hold mirrors with two hands. Walk, don't run, in the dark.
- [ ] Only the Team Leader calls directions: left a little, stop, hold.
- [ ] Write every trial down, even the bad ones. Bad trials teach the most.
- [ ] Change one thing at a time so you know what worked.

My build kit

- [ ] This logbook and 2 pencils
- [ ] 1 protractor (the only tool allowed at the test)
- [ ] A flashlight
- [ ] 2 to 4 small mirrors (or foil smoothed on cardboard)
- [ ] Painter's tape and a paper target
- [ ] A stopwatch or timer

Logbook map

Know your event	p. 3-4
Path plans (graph paper)	p. 5-6
Trial Logs	p. 7-10
Best-time graph	p. 11
What went wrong?	p. 12
At-home build challenges	p. 13-16
Tournament checklist	p. 17
Missions and badges	p. 18-19

Where to practice

A dark room
any room with blinds and a switch: your practice lab

Kitchen
shiny spoons, foil, a glass of water that bends light

Bathroom mirror
watch your pupils get big and small

Windows
which things let light through: transparent, translucent, opaque?

Car or bus
side mirrors and headlight beams in fog

Open Lab Fridays
the Beam Me Up bin has mirrors and a flashlight

Did you know? Light is the fastest thing in the universe. A beam could zip around the Earth more than 7 times in one second, so your 60-second challenge is all about your hands, not the light!

Know your event: the rules

Teams of 3 take a 50-minute written test on light and the eye, and are called out one team at a time to a darkened room to bounce a fixed flashlight beam off 3 handheld mirrors plus 1 wall mirror onto a target, twice, in 60 seconds or less each (bring only pencils and 1 protractor).

Part	What the rules say
Team	3 students. One is the Team Leader, who tells the judge when you are ready.
Written test	Questions about light and the parts of the eye.
Challenge	Use 3 handheld mirrors and 1 wall mirror to send a flashlight beam onto a target.
Time	1 minute to plan with the lights on. Up to 60 seconds in the dark. The beam must stay on the target for 3 seconds.
The course	The flashlight is set up by the judges. You may not touch it. The beam must hit all 3 handheld mirrors and the wall mirror, then the target.
Taped box	A 160 x 80 cm box is taped on the floor in front of the wall mirror. Mirror holders stay outside it.
Two runs	You run the challenge twice. Three timers time each run and the middle time counts.
What to bring	Only pencils and 1 protractor. No flashlights, mirrors, timers, phones, or smartwatches.

How it's scored

- **The written test is the base score.** Don't rush it after the challenge.
- The challenge adds **bonus points**, but only for the **fastest 5 teams**.
- The beam must stay on the target for **3 full seconds**. After 60 seconds, time is up.
- Practice score: average your two course times (lower is better).

Team jobs

Team Leader: tells the judge "Ready" and is the only one who calls directions.

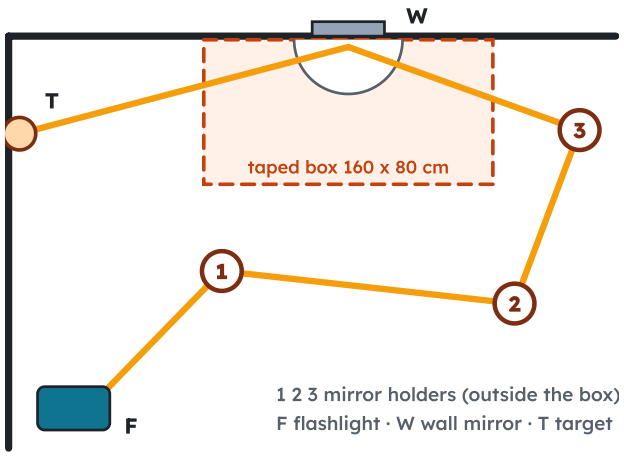
Mirror 1: catches the flashlight beam first, then freezes.

Mirror 2 and 3: pass the beam on to the wall mirror or the target.

Challenge strategy (from Lesson 3)

1. Lights on: find the flashlight, the wall mirror, the target, and anything in the way.
2. Decide the order the beam will travel and where each mirror holder stands (outside the taped box).
3. Use angle in equals angle out at the wall mirror to find where to stand.
4. Team Leader says Ready. In the dark, Mirror 1 finds the beam and freezes. Then Mirror 2, then Mirror 3.
5. Tiny moves only. Hold the spot on the target for 3 seconds.

One way the course can look (from above)

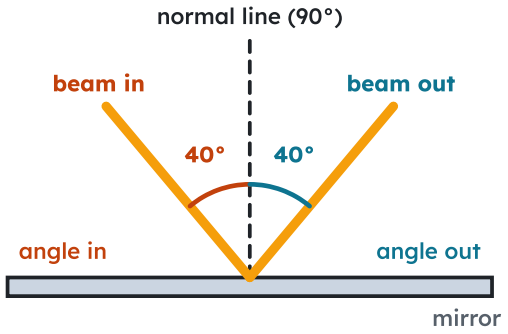


Know your science: light and the eye

The five light words

Reflection	Light bounces off a surface. Smooth, shiny things reflect best.
Refraction	Light bends when it goes from one material into another, like air into water.
Absorption	Light soaks into a material and turns into heat. Dark colors absorb the most.
Transmission	Light passes through a material.
Scattering	Light bounces off tiny bits (dust, water drops, air) in many directions. Air scatters blue light most, so the sky looks blue.
Transparent / Translucent / Opaque	Clear, light passes through / Some blurry light passes through / No light passes through.

Law of reflection



The angle in **equals** the angle out. Both are measured from the **normal line**, which sticks straight out of the mirror. Know the angle from the mirror's surface? Subtract it from 90.

The 11 eye parts (in the order light travels, more or less)

Conjunctiva	Thin, clear skin that covers the white of the eye and the inside of the eyelids. Keeps the eye moist and protected.
Cornea	Clear dome on the front of the eye. Light enters here. It does most of the bending (refraction) of light.
Sclera	The tough white outer coat. Protects the eye and keeps its shape.
Iris	The colored ring. It is a muscle that makes the pupil bigger or smaller.
Pupil	The black hole in the middle of the iris. Lets light into the eye.
Lens	Clear, bendy part behind the pupil. Changes shape to focus light on the retina.
Ciliary muscle	Ring of muscle around the lens. Pulls to change the lens shape so we can focus near or far.
Vitreous humor	Clear jelly that fills the inside of the eye and keeps it round. Light passes through it.
Retina	Thin layer at the back of the eye where the picture lands (upside down).
Photoreceptors: rods	Light-catching cells in the retina. Work in dim light. See black, white, and gray.
Photoreceptors: cones	Light-catching cells in the retina. See color and fine detail. Need bright light.
Optic nerve	Carries signals from the retina to the brain. The brain makes the picture and flips it right side up.

Light's path: cornea → pupil → lens → vitreous humor → retina (rods and cones) → optic nerve → brain. The picture lands upside down, and the brain flips it.

Path Plan A: Red course: big target

Draw the room from above (a bird's-eye view). Mark the flashlight (F), the wall mirror (W), the target (T), the taped box, furniture in the way, and where Mirror 1, 2, and 3 stand. Draw the beam with a ruler and write each angle from the normal.

Design name _____ **Date** _____ **Version #** _____

Bird's-eye view of the room

1 small square = ____ cm

Beam order (F, 1, 2, 3, W, T or another order):

Why I think this path is fastest:

Path Plan B: Blue course: small target

Draw the room from above (a bird's-eye view). Mark the flashlight (F), the wall mirror (W), the target (T), the taped box, furniture in the way, and where Mirror 1, 2, and 3 stand. Draw the beam with a ruler and write each angle from the normal.

Design name _____ **Date** _____ **Version #** _____

Bird's-eye view of the room

1 small square = ____ cm

Beam order (F, 1, 2, 3, W, T or another order):

Why I think this path is fastest:

Trial Log #1

Every run is a trial. Fill a row right after each one. Result: the middle time of 3 timers, and Y if the spot held on the target for 3 seconds.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (time in sec)	Result (sec, held 3 s? Y/N)	What I learned
1				
2				
3				
4				
5				
6				
7				
8				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #2

Every run is a trial. Fill a row right after each one. Result: the middle time of 3 timers, and Y if the spot held on the target for 3 seconds.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (time in sec)	Result (sec, held 3 s? Y/N)	What I learned
9				
10				
11				
12				
13				
14				
15				
16				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #3

Every run is a trial. Fill a row right after each one. Result: the middle time of 3 timers, and Y if the spot held on the target for 3 seconds.

Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (time in sec)	Result (sec, held 3 s? Y/N)	What I learned
17				
18				
19				
20				
21				
22				
23				
24				

Best trial on this page: # _____ **Why it worked:** _____

Trial Log #4

Every run is a trial. Fill a row right after each one. Result: the middle time of 3 timers, and Y if the spot held on the target for 3 seconds.

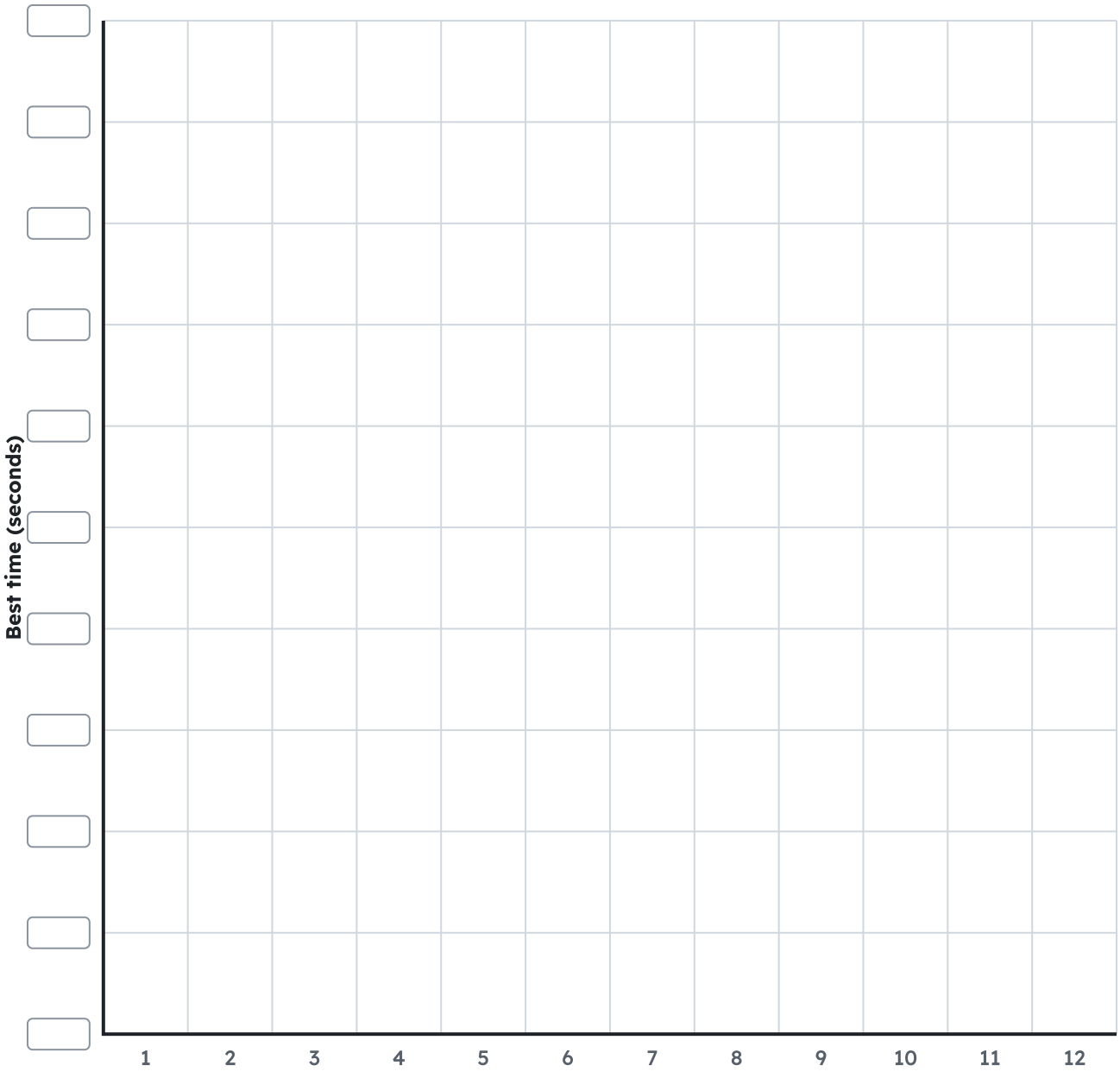
Date _____ **Design or setup** _____ **Teammates** _____

Trial	What I changed (change 1 thing!)	My prediction (time in sec)	Result (sec, held 3 s? Y/N)	What I learned
25				
26				
27				
28				
29				
30				
31				
32				

Best trial on this page: # _____ **Why it worked:** _____

My best-time graph

After each practice day, put a dot for your best challenge time (the middle of 3 timers). Write the numbers on the side (count by 5s or 10s), then connect the dots.



Date of #	1	2	3	4	5	6	7	8	9	10	11	12

Lower is better! Draw a red line across at 60 seconds: under it, you beat the clock. Every dot under the line counts toward the Under-60 Club badge.

My best so far: _____

My tournament goal: _____

Is my line going the right way? _____

What went wrong?

Every engineer's design fails sometimes. Good engineers figure out **why**. Here are problems Beam Me Up teams hit a lot, and fixes to try.

Problem	What it looks like	Try this (one fix at a time)
Lost the beam in the dark	The spot disappears and everyone waves mirrors around.	Mirror 1 finds the beam first and freezes. Then Mirror 2 finds it, then Mirror 3.
Spot jumps past the target	Tiny tilts make the spot fly across the wall.	Two hands, tiny turns. Far-away targets need the smallest moves.
Everyone talking	Three people call directions at once.	Only the Team Leader calls: left a little, up a little, stop, hold.
Beam blocked	A body or a head is in someone's beam.	Plan where each person stands so nobody steps in front of a beam.
Stopped too soon	The spot touched the target, then slid off.	Hold the spot for 3 full seconds. Count 1-1000, 2-1000, 3-1000.
Inside the taped box	A mirror holder steps over the tape.	Stand outside the box. Use angle in = angle out to find your spot.
Ran out of plan time	Still arguing when the lights go off.	Use the same plan steps every time: find F, W, T, then the order.

My fix-it log

Write down each problem you hit. Fix 3 problems to earn the **Troubleshooter** badge.

What went wrong	Why I think it happened	What I tried	Did it work?
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no
			☐ yes ☐ no

AT-HOME BUILD CHALLENGE 1 · AFTER LESSON 1 (FRI OCT 23)

Mirror Maze



Beam Bouncer badge

Your mission: bounce a flashlight beam off **2 mirrors** onto a paper target, and hold it there for 3 seconds.

You need

- [ ] A flashlight
- [ ] 2 small mirrors (or foil smoothed flat on cardboard)
- [ ] Books to stand the flashlight on
- [ ] Tape and a paper target (draw a circle)
- [ ] A timer and a helper

Build it

1. Tape the target on a wall.
2. Set the flashlight on a stack of books. Now don't touch it (like the real event).
3. Mirror 1 catches the beam and sends it to Mirror 2.
4. Mirror 2 sends the beam to the target. Tiny turns!
5. Time 3 tries. Then move the target and try again.

Draw your maze from above: flashlight, mirrors, target, and the beam

My results

Try	Where was the target?	Time (sec)	Held 3 sec? (Y/N)
1			
2			
3			

What happened to the spot on the wall when you tilted a mirror just a tiny bit?

Safety: never shine the flashlight or a mirror spot into anyone's eyes. Ask an adult before you dim the lights.

Date done: _____ **Coach or Captain initials (Beam Bouncer badge):** _____

AT-HOME BUILD CHALLENGE 2 · AFTER LESSON 2 (FRI DEC 4)

Periscope

Periscope Pro badge

Your mission: build a periscope with 2 mirrors so you can peek over a couch. It works because angle in = angle out.

You need

- [ ] An empty milk or juice carton (half-gallon size), rinsed
- [ ] 2 small mirrors about 2 x 2 in (craft or makeup mirrors)
- [ ] Tape and a ruler
- [ ] Scissors (an adult does the cutting)
- [ ] Your protractor

Build it

1. An adult cuts a square window near the top of one side, and one near the bottom of the **opposite** side.
2. Tape Mirror 1 inside the top at a 45 degree slant, shiny side facing the top window and down the tube.
3. Tape Mirror 2 inside the bottom at 45 degrees, shiny side facing up the tube and out the bottom window.
4. Look in the bottom window. Peek over a couch or around a door.
5. Check each slant with your protractor. Adjust until the view is clear.

Draw a side view: both mirrors, the normal lines, and the light's path

My results

What I looked over or around	Could I see it? (Y/N)	What I fixed

Light hits each mirror 45 degrees from the normal. What is the angle of reflection? Why does the light come out going the same way it went in?

Safety: an adult does all the cutting. Mirrors can have sharp edges, so tape them before you handle them.

Date done: _____ **Coach or Captain initials (Periscope Pro badge):** _____

AT-HOME BUILD CHALLENGE 3 · AFTER LESSON 3 (FRI JAN 22)

Pinhole Eye

Eye Engineer badge

Your mission: build a model eye that makes an upside-down picture, just like the picture on your retina.

You need

- [ ] A paper towel tube
- [ ] Foil and wax paper (about 4 x 4 in each)
- [ ] 2 rubber bands or tape
- [ ] A pin or a sharp pencil (an adult pokes the hole)
- [ ] A dark towel or a blanket

Build it

1. Cover one end of the tube with foil. An adult pokes a tiny hole in the middle. This is the **pupil**.
2. Cover the other end with wax paper, pulled tight. This is the **retina** screen.
3. Go to a dim room. Put the towel over your head and the wax paper end.
4. Point the foil end at a bright window or a lamp. **Never at the sun.**
5. Look at the wax paper. Draw what you see.

Draw what you saw on the wax paper screen

My results

What I pointed at	What the picture looked like	Upside down? (Y/N)

Which eye part is like the wax paper? Which part of you flips the picture right side up?

Safety: never look at the sun, even through your pinhole eye. An adult pokes the hole.

Date done: _____ **Coach or Captain initials (Eye Engineer badge):** _____

AT-HOME BUILD CHALLENGE 4 · AFTER LESSON 4 (FRI FEB 19)

Living-room 4-Mirror Course

Beam Captain badge

Your mission: set up the real challenge at home and run it as the **Team Leader**: 1 minute to plan, lights off, 60 seconds to hit the target.

You need

- [ ] A flashlight on a stack of books
- [ ] 3 mirrors, plus 1 propped up as the wall mirror
- [ ] Painter's tape for a keep-out box
- [ ] A paper target
- [ ] A timer and 2 or 3 family helpers

Build it

1. Set up the flashlight, the wall mirror, and the target like the real event.
2. Tape a keep-out box on the floor in front of the wall mirror.
3. Lights on: plan for 1 minute. Say **Ready**.
4. Lights off (ask an adult first). Count 3-2-1, go. You call the moves: left a little, stop, hold.
5. Hold the spot 3 seconds. Time 3 runs.

Bird's-eye view of your living-room course

My results

Run	Plan time (sec)	Challenge time (sec)	Held 3 sec? (Y/N)
1			
2			
3			

Which calls worked best for your helpers? What will you say at the tournament?

Safety: keep the floor clear so nobody trips in the dark, and never shine light in anyone's eyes.

Date done: _____ **Coach or Captain initials (Beam Captain badge):** _____

Tournament-day checklist

Sat Mar 13, Chatham Regional (Seaforth High School). Check each box as you go.

Pack your bag

- [ ] 2 sharp pencils
- [ ] 1 protractor (the **only** tool allowed)
- [ ] This logbook to review on the way (it stays with your coach during the event)
- [ ] A water bottle and a snack
- [ ] Your 19 flashcards to review while you wait

The night before

- [ ] Quiz all 19 flashcards: 8 light words and 11 eye parts.
- [ ] Say the law of reflection out loud.
- [ ] Decide who is Team Leader and Mirror 1, 2, and 3.
- [ ] Get a good night's sleep.

Rules to remember

- No light sources, mirrors, timers, phones, or smartwatches.
- Stay outside the taped box.
- Don't touch the flashlight.
- 60 seconds max per run.

At the event

- [ ] Start the written test right away: it is the base score.
- [ ] When the judge calls you, stop the test and go to the challenge room.
- [ ] Lights on: find F, W, and T, and anything in the way. Plan in 1 minute.
- [ ] Team Leader says **Ready**. Mirror 1 finds the beam and freezes, then 2, then 3.
- [ ] Tiny moves. Hold the spot 3 seconds. Then go back and finish the test.

After: my tournament report

Our score or time:

What went great:

What we would change:

Thank you to:

Draw or write my favorite moment of the day

Build Missions

One mission for each Beam Me Up meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge on page 19 and in your Team Badge Passport.

When	Mission	Badge	Done	Initials
After Fri Oct 23 Lesson 1	Build the Mirror Maze: 2 bounces onto a target (p. 13).	Beam Bouncer	☐	
After Fri Dec 4 Lesson 2	Build a periscope and explain its 45 degree angles (p. 14).	Periscope Pro	☐	
After Fri Jan 22 Lesson 3	Build a Pinhole Eye and see the upside-down picture (p. 15).	Eye Engineer	☐	
After Fri Feb 19 Lesson 4	Run the living-room 4-mirror course as Team Leader (p. 16).	Beam Captain	☐	
Anytime Plan	Plan both mirror paths on graph paper (p. 5-6).	Path Planner	☐	
Anytime Log	Log 12 trials (p. 7-10) and graph your best times (p. 11).	Trial Tracker, Graph Maker	☐	
Anytime Bonus	Fix 3 problems (p. 12). Finish a course in under 60 seconds.	Troubleshooter, Under-60 Club	☐	

Tournament day: Sat Mar 13, Chatham Regional (Seaforth High School)

- ☐ 2 pencils
- ☐ 1 protractor
- ☐ Your 19 flashcards

- Full checklist on page 17.
- The written test is the base score. Finish it!
- Plan in 1 minute, then tiny moves in the dark.

My Badges

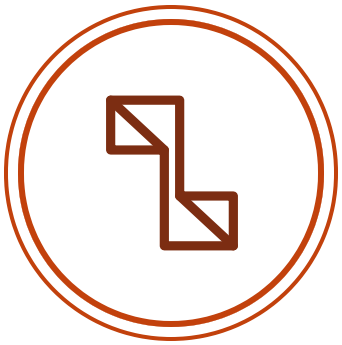
Color in each badge when you earn it. Earn all 9 to become a **Master Beam Engineer**! Your Team Badge Passport has the same badges.



Beam Bouncer

Mirror Maze: 2 bounces onto a target

Initials: _____



Periscope Pro

Build a periscope and explain its angles

Initials: _____



Eye Engineer

Pinhole eye: see the upside-down picture

Initials: _____



Beam Captain

Lead the 4-mirror course at home

Initials: _____



Path Planner

Plan 2 mirror paths on graph paper

Initials: _____



Trial Tracker

Log 12 trials in the Trial Log

Initials: _____



Graph Maker

Graph your best challenge times

Initials: _____



Troubleshooter

Fix 3 problems on What went wrong?

Initials: _____



Under-60 Club

Hit the target and hold 3 sec in under 60 sec

Initials: _____